Partow Roshani

Computer Science Undergraduate

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Profile

Computer Science undergraduate at Shahid Beheshti University with interests in **Machine Learning, Data Science, Digital Image Processing, Algorithms, and Software Development**. Experienced in Python, C++, and Java through academic and applied projects spanning predictive maintenance, deep learning, graph algorithms, concurrent programming, and client–server software. Teaching Assistant with experience in programming instruction, assignment design, individual mentoring, and software project supervision.

Education

Shahid Beheshti University Tehran, Iran Sep. 2023 – Present

B.Sc. in Computer Science

- **Cumulative GPA: 18.95/20**
- Relevant coursework: Machine Learning, Data Science, Digital Image Processing, Data Structures & Algorithms, Concurrent Programming, Advanced Programming, and Basic Programming.
- Selected grades: Data Science **20/20**; Machine Learning **19.75/20** Digital Image Processing **20/20**; Concurrent Programming **20/20**; Computer Systems Fundamentals **20/20**; Digital Logic Laboratory **20/20**; Advanced Programming **20/20**; Basic Programming **19.75/20**.

Alghadir High School Kish, Iran Jul. 2023

Diploma in Mathematics and Physics

- Graduated with GPA **19.84/20**.

Teaching Experience

Shahid Beheshti University Tehran, Iran Feb. 2026 – Aug. 2026

Teaching Assistant – Advanced Programming

- Served as a Teaching Assistant for the **Advanced Programming** course under **Dr. Saeed Reza Kheradpisheh**
- Taught **Object-Oriented Programming (OOP)** concepts and supported students in applying object-oriented design principles in Java.
- Designed a programming assignment focused on **OOP concepts**, including class design, encapsulation, inheritance, polymorphism, and software structure.
- Mentored **three students individually**, providing regular support with programming concepts, debugging, problem solving, and code organization.
- Mentored **two student teams during their final projects**, advising them on software architecture, implementation decisions, debugging, and overall project development.

Shahid Beheshti University Tehran, Iran Sep. 2025 – Feb. 2026

Teaching Assistant – Basic Programming

- Served as a Teaching Assistant for **Dr. Saeed Reza Kheradpisheh** in the **Basic Programming** course.
- Supported first-year Computer Science students in fundamental programming and problem-solving techniques.
- Taught and assisted with core programming topics including **data types, control flow, loops, and functions**.
- Designed and reviewed programming exercises and helped students improve debugging, algorithmic thinking, and code organization.

Alghadir High School Kish, Iran Nov. 2021 – Jan. 2022

Teaching Assistant – Physics and Chemistry

- Assisted students with coursework and problem solving in Physics and Chemistry.

Selected Projects

Jet Engine Hospital – Predictive Maintenance

Machine Learning

- Built an end-to-end predictive-maintenance system using **NASA C-MAPSS FD001, FD003, and FD004** turbofan-engine datasets.
- Developed pipelines for **Remaining Useful Life (RUL) regression**, failure-risk classification at 10, 20, and 30-cycle horizons, anomaly detection, and uncertainty estimation.
- Applied leakage-safe engine-level splitting, temporal feature engineering, operating-condition-aware processing,

probability calibration, and conformal prediction intervals.
- Integrated model outputs into maintenance decision policies and developed an interactive **Streamlit dashboard** for engine-level inference and visualization.
- **Tech:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Streamlit.

Deep Image Restoration and Classification — Digital Image Processing / Deep Learning

- Developed an end-to-end deep-learning pipeline for image denoising, deblurring, model comparison, and semantic image classification.
- Evaluated **CNN, ResNet, and U-Net** architectures for image deblurring using PSNR and automatically selected the best-performing model.
- Trained a **PreActResNet18** classifier on CIFAR-10, achieving **87.11% test accuracy** and **0.8897 validation macro F1-score**.
- Worked with both **TensorFlow and PyTorch** within the same image-processing pipeline.
- **Tech:** Python, PyTorch, TensorFlow, NumPy, Matplotlib.

Personalized PageRank for Fraud Detection — Data Structures & Algorithms

- Implemented a Personalized PageRank framework for large-scale graph analysis using a performance-oriented **C++17 backend**.
- Implemented Power Iteration, deterministic seed selection, convergence monitoring, Precision@K evaluation, and scalability experiments.
- Designed batch experiments across datasets and algorithm parameters and generated structured results for comparative analysis.
- Built a Python/Tkinter graphical interface that communicates with the C++ backend through structured JSON output.
- **Tech:** C++, Python, Graph Algorithms, Tkinter, JSON, CSV.

Telegram Clone — Advanced Programming

- Developed a real-time messaging application using a client–server architecture with **Java, JavaFX, PostgreSQL, and socket programming**.
- Implemented multithreaded server-side connection handling and structured JSON communication between clients and the server.
- Supported private chats, saved messages, groups, channels, message persistence, and delivery/read states.
- Worked on GUI development, database integration, concurrency, application logic, and client–server communication.
- **Tech:** Java, JavaFX, PostgreSQL, Socket Programming, Multithreading, JSON.

Technical Skills

Programming Languages:	Python, C++, Java, SQL, MATLAB
Machine Learning / Data:	Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, CatBoost, SHAP
Visualization:	Matplotlib, Plotly
Software Development:	Object-Oriented Programming, Data Structures & Algorithms, Concurrent Programming, Multithreading, Socket Programming
Databases:	PostgreSQL
Tools:	Git, GitHub, Jupyter Notebook, VS Code, Streamlit, JavaFX, Scene Builder

Achievements

Ranked 1st among the 2023 Computer Science undergraduate cohort at Shahid Beheshti University, based on academic performance through the end of sixth semester. — Iran 2026

Rank **1**/50

Iran National University Entrance Examination (Konkur): Regional Rank **198**/41,279 — Iran 2023
National Rank **2,742**/147,317

Languages

Persian: Native

English: Intermediate – IELTS Academic **5.5**
2023